

SPORTS INJURY RISK DETECTION FROM VIDEO

Implementation-Justified Project Documentation

MotionGuard AI

Computer Vision • Pose Estimation • Biomechanical Analysis • Machine Learning

Internship Report

Academic Year 2026–2027

1. PROJECT OVERVIEW

Sports Injury Risk Detection from Video is an AI-assisted movement-analysis system that converts ordinary athlete video into measurable movement information and a model-based injury-risk estimate. The implemented interface, shown in the supplied screenshots, demonstrates the complete user-facing workflow: landing page, authentication, athlete dashboard, movement-video upload, raw-video preview, real-time pose skeleton, biomechanical indicators, athlete context, physical-condition inputs, and pipeline readiness.

The implementation is therefore more than a static prediction screen. It establishes a complete analysis workflow in which video is the primary source of movement information, while athlete profile and contextual information can be used to personalize the feature-processing and prediction stages.

2. JUSTIFICATION OF THE IMPLEMENTED SYSTEM

Each major screen and metric visible in the implementation corresponds to a specific stage of the proposed AI pipeline. The following sections justify why each element is included and how it contributes to the project objective.

2.1 Landing Page — Movement Intelligence

The landing page communicates the central purpose of the system: understanding movement and using the resulting measurements to estimate risk. The displayed examples—knee angle, symmetry, movement score, and risk—are not arbitrary dashboard decorations. They represent the types of outputs produced after pose landmarks have been converted into biomechanical indicators and passed through the prediction workflow.

The visual separation between movement understanding and risk prediction reflects the project's two-stage logic: first measure movement, then use those measurements for risk estimation.

2.2 Login and Role-Aware Access

The login screen provides access control for the application. The presence of athlete, coach, and administrator demo roles reflects the intended multi-user workflow. Athlete users can submit their own movement analyses, coaches can review athlete-level results, and administrative functionality can be used for system or account management. Authentication also provides a foundation for protecting uploaded movement videos and analysis records.

2.3 Athlete Dashboard

The dashboard summarizes the athlete's analysis history and current context. The cards for Latest Risk, Average Risk, Processed Videos, and Model Runs are operational indicators rather than isolated statistics. Latest Risk represents the most recent model output, Average Risk can summarize historical assessments, Processed Videos measures usage of the video pipeline, and Model Runs records model-analysis executions.

The Athlete Profile & Physical Metrics section contains height, weight, BMI, primary sport, position, and coach affiliation. These fields provide contextual information that can be used when the trained model requires additional athlete-specific inputs. The interface also explicitly distinguishes unavailable information from populated information, which is important because the system should not silently invent missing inputs.

2.4 New Analysis — Video Upload

The New Analysis screen is the main entry point to the computer-vision pipeline. It accepts common video formats and communicates that full-body framing is preferred for pose tracking. This is technically justified because missing body parts, occlusion, poor framing, or very small subjects can reduce landmark quality and consequently degrade biomechanical feature calculations.

The Raw Video preview confirms that the uploaded file has been successfully received before analysis. This reduces user error and allows the user to verify that the selected movement is suitable for analysis.

2.5 Athlete Context

The Athlete Context panel shows which contextual variables are available for the analysis. The screenshot demonstrates states such as Profile: Missing, Sport: Not provided, and BMI/Training duration: Fallback median. This is an important implementation detail: when personalized information is unavailable, the system uses a documented fallback rather than pretending that a personalized value exists.

Using a training-set median as a fallback can keep the pipeline operational, but it must be clearly labeled because the resulting prediction is less personalized than one based on actual athlete data.

2.6 Current Physical Condition

The Pain Level and Pain Location inputs provide additional context for interpretation. These fields should not be treated as model-generated diagnoses. Their purpose is to allow the athlete to provide

relevant current-condition information that can be reviewed alongside the AI result. A clinical decision should remain with an appropriately qualified professional.

3. JUSTIFICATION OF COMPUTER VISION OUTPUT

3.1 MediaPipe Pose Estimation

The Real-Time Skeleton screen explicitly identifies MediaPipe Pose Estimation and displays a 33-point body skeleton. This directly supports the project's pose-estimation stage. The body landmarks provide the coordinates needed to calculate joint angles, segment orientation, symmetry, range of motion, and temporal movement characteristics.

The screenshot also displays FPS and a frame number, demonstrating that pose processing is performed frame by frame over a video sequence rather than on a single still image.

3.2 Knee Angle

The knee angle shown in the implementation is calculated from the relative positions of the hip, knee, and ankle landmarks. This is a meaningful biomechanical feature because knee configuration changes throughout running, squatting, jumping, and landing movements. The implementation displays both left/right knee measurements where available, making the output suitable for bilateral comparison.

The displayed value should be interpreted as an image-derived movement measurement. Camera angle, landmark confidence, occlusion, and perspective can affect the measurement.

3.3 Keypoint Tracking

The interface reports 33 landmarks and applies a visibility/confidence threshold. This provides a quality-control step before feature calculations. A confidence threshold helps prevent low-quality landmarks from being treated as equally reliable measurements.

3.4 Temporal Smoothing

The implementation displays an EMA smoothing parameter, indicating that landmark measurements are temporally filtered. Exponential moving average smoothing reduces frame-to-frame jitter and produces more stable joint-angle and movement curves. The trade-off is that excessive smoothing can remove rapid movement changes, so the smoothing parameter must be validated for the selected sport and movement.

3.5 Cadence and Gait

The Cadence & Gait panel indicates that ankle movement is used to identify movement cycles. For running analysis, cadence provides temporal information about steps and can complement joint-angle and gait-speed measurements.

3.6 Movement Type and Camera View

The Movement Type selector identifies the activity being analyzed, such as Running. The Camera View selector identifies the viewpoint, such as Side. These inputs are technically important because the reliability and interpretation of biomechanical features depend on the movement and camera geometry. A side view is particularly useful for sagittal-plane measurements such as knee flexion and trunk lean, while other views may be more appropriate for lateral alignment or symmetry analysis.

4. JUSTIFICATION OF THE AI PREDICTION

The AI model receives structured features derived from the video-processing pipeline. Candidate features include joint-angle statistics, gait speed, cadence, step count, jump height, range of motion, symmetry, and contextual training variables. The purpose of the prediction model is to identify patterns associated with the target injury outcome in the training data.

The risk label displayed in the interface should be described as an estimated model output. It should not be presented as proof that an athlete is injured or will become injured. The appropriate project claim is that the system provides AI-assisted injury-risk screening based on movement and contextual features.

4.1 Movement Score

The Movement Score is a presentation-level summary intended to combine selected movement-quality indicators into an easy-to-understand score. It should be computed from defined normalized metrics rather than manually assigned. The report should document the exact formula used by the implementation if the score is included in the final evaluation.

4.2 Symmetry Score

The symmetry percentage summarizes the similarity between corresponding left and right movement measurements. For example, left and right knee-angle excursions can be compared over a movement cycle. A high similarity percentage indicates that the selected measurements are close to one another; it does not by itself establish that the athlete is injury-free.

4.3 Risk Level

The Low/Moderate/High category is a communication layer placed on top of the model probability. The thresholds should be determined using validation data and documented in the final implementation. The category is intended for screening and visualization and should not be represented as a medical classification.

5. DATASET AND MODEL JUSTIFICATION

The project's structured dataset contains 15,420 records with 29 columns and uses injury_occurred as the binary target. The target distribution contains 9,868 non-injury records and 5,552 injury records. This distribution means that evaluation must include class-sensitive metrics rather than relying on accuracy alone.

Dataset Element	Role in Project
joint_angles	Core biomechanical movement feature
gait_speed	Movement-speed feature
cadence	Temporal gait feature
step_count	Activity/gait volume feature
jump_height	Explosive-movement feature
range_of_motion	Joint mobility feature
training_intensity	Training-load context
training_duration	Exposure-duration context
injury_occurred	Binary prediction target
athlete_id / session_id	Identifiers removed from core feature set

6. END-TO-END SYSTEM WORKFLOW

1. User authenticates and opens the analysis dashboard.
2. User creates a new analysis and uploads a movement video.
3. System validates and previews the selected video.
4. Video frames are processed sequentially.
5. MediaPipe Pose estimates 33 body landmarks.

6. Landmarks below the configured visibility/confidence requirement are filtered.
7. Temporal smoothing reduces keypoint jitter.
8. Joint angles, gait/cadence, symmetry, range of motion and related features are calculated.
9. Movement/context features are passed through the trained preprocessing pipeline.
10. The machine-learning model generates an injury-risk estimate.
11. Explainability information can be generated from the model inputs.
12. Dashboard displays movement metrics, risk category, and supporting information.

7. IMPLEMENTATION EVIDENCE FROM THE PROVIDED OUTPUT

The supplied screenshots provide direct evidence for the following implemented components:

Observed Output	What It Demonstrates
MotionGuard AI landing page	Project identity and movement/risk analysis concept
Login screen	Authentication and role-aware access concept
Dashboard	Analysis history, risk summary, profile and metrics
New Analysis screen	Video ingestion and analysis setup
Raw video preview	Successful video selection and playback
MediaPipe Pose / 33 body keypoints	Pose-estimation stage
Knee: 59 deg / R-Knee: 143 deg	Live joint-angle calculation
FPS: 25 / frame number	Frame-by-frame temporal processing
33 Landmarks / visibility > 0.40	Keypoint quality filtering
EMA $\alpha = 0.3$	Temporal smoothing
Cadence & Gait / pelvis waveform	Temporal movement analysis
Running / Side selectors	Movement and camera-context controls
Athlete Context / fallback median	Handling of missing model inputs
Pipeline Readiness checklist	Pre-analysis validation state

8. RESULTS AND DISCUSSION

The screenshots demonstrate that the project has progressed from a conceptual injury-risk model to an integrated application workflow. In particular, the Real-Time Skeleton output proves that the system is extracting body landmarks from an actual movement video and calculating dynamic joint measurements. The presence of frame number and FPS indicates sequence-based processing, while the EMA filter and visibility threshold demonstrate explicit signal-quality handling.

The dashboard further demonstrates how raw model outputs are intended to be converted into user-facing information. Rather than exposing only a numerical classifier output, the interface provides movement metrics, athlete context, and analysis status. This makes the system more suitable for demonstration and review by users who may not be familiar with machine-learning internals.

The current output should nevertheless be described honestly: the screenshots establish functional pipeline behavior and UI integration, but they do not by themselves establish clinical validity or model generalization. Final model performance must be supported by a held-out evaluation set and documented metrics such as precision, recall, F1-score, ROC-AUC, PR-AUC, and a confusion matrix.

9. LIMITATIONS

- Pose quality depends on camera viewpoint, lighting, framing, occlusion, and video resolution.
- Monocular video does not directly provide full laboratory-grade 3D biomechanics.
- Fallback median values reduce personalization when athlete context is missing.
- Movement scores and risk thresholds require documented formulas and validation.
- An AI association is not evidence of causation.
- The system is a screening/decision-support tool and not a medical diagnostic device.

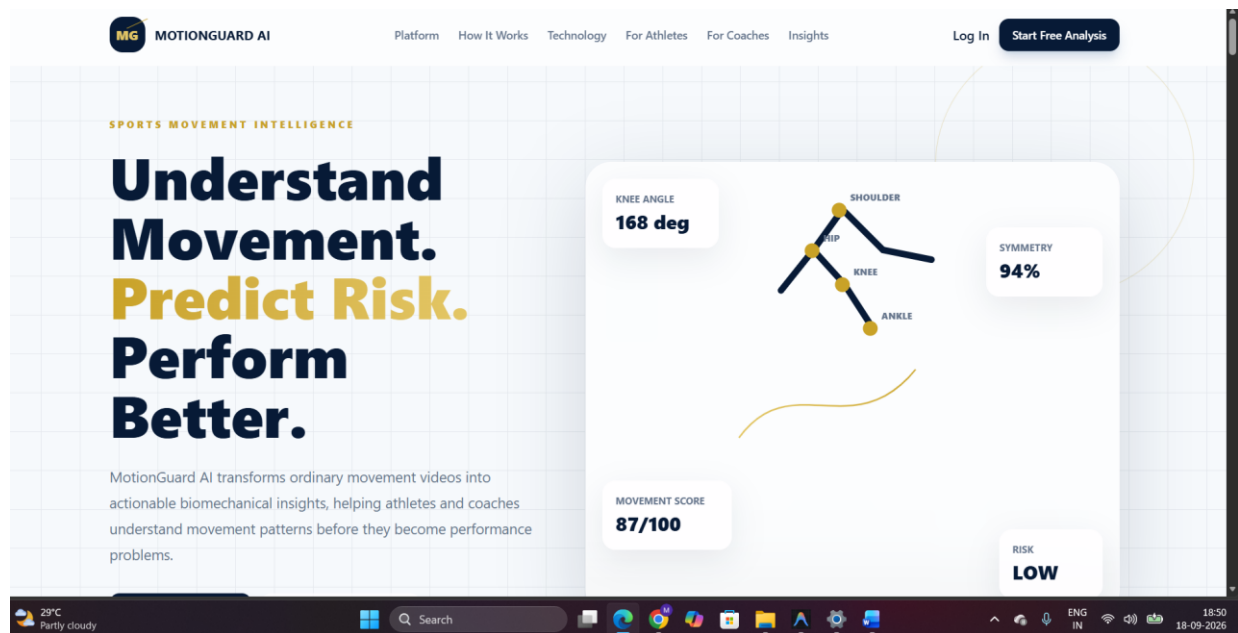
10. CONCLUSION

The implemented MotionGuard AI application provides a coherent end-to-end demonstration of sports injury-risk analysis from video. The supplied outputs justify the use of pose estimation, dynamic knee-angle measurement, bilateral comparison, cadence/gait analysis, keypoint confidence filtering, temporal smoothing, athlete context, and machine-learning-based risk presentation.

The strongest technical evidence in the implementation is the real-time 33-landmark skeleton with frame/FPS information and dynamic knee-angle values. These outputs show that the application is extracting quantitative movement information rather than merely displaying a pre-written risk label. The final academic evaluation should combine this implementation evidence with measured model-performance results and clearly state the limitations of video-based risk estimation.

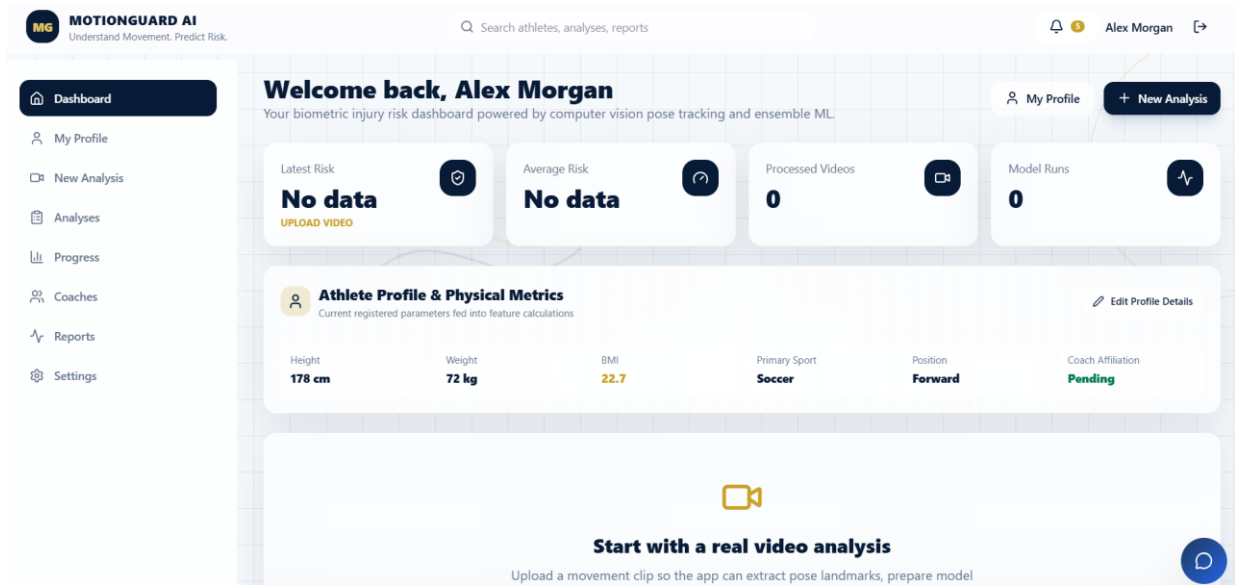
11. APPENDIX — SCREENSHOT EVIDENCE

Figure 1. Implementation screenshot



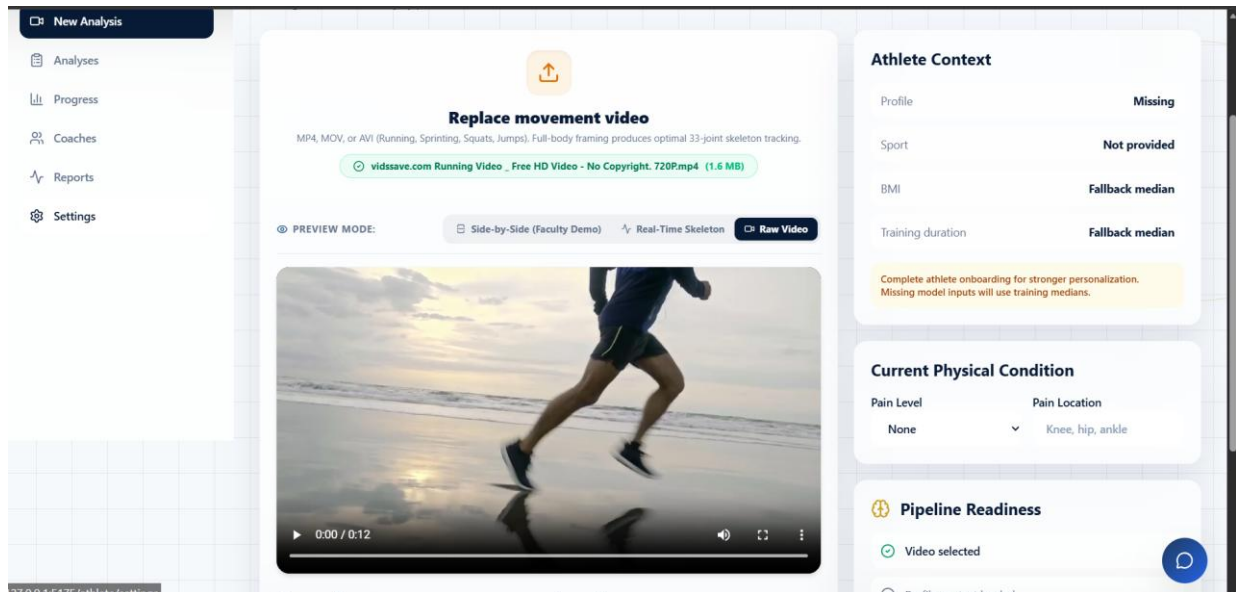
Landing page showing the MotionGuard AI concept and example movement/risk indicators.

Figure 3. Implementation screenshot



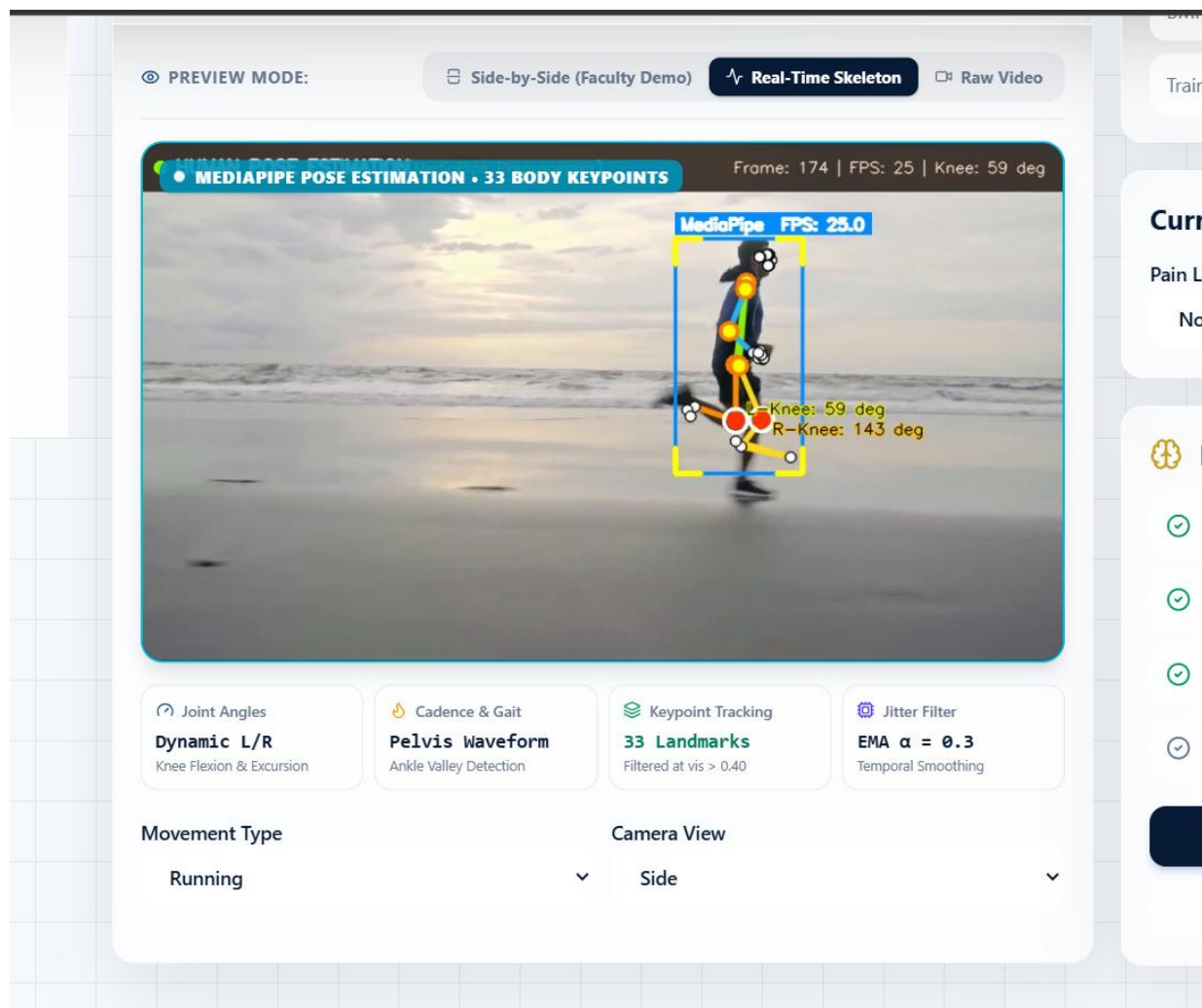
Athlete dashboard showing risk, video, model-run and physical-profile sections.

Figure 4. Implementation screenshot



New Analysis interface showing uploaded movement video, athlete context and pipeline readiness.

Figure 5. Implementation screenshot



Real-time pose-analysis output showing 33 landmarks, dynamic knee-angle measurements, cadence/gait processing, visibility filtering and EMA smoothing.